Meeting 3. Probabilities and the Sigmoid

Act 1 Uncertainty in Measurements · Act 2 The Newark Weather Dataset · Act 3 The Sigmoid Function · Act 4 Interpreting Classification Errors

About these notes. These instructor-written notes are the course text for CS 301 and are posted on Canvas after each meeting. They include examples and questions discussed in class. The exams cover these notes and your assignment notebook. See the syllabus for the schedule, grading, and other course information.

Meetings 1 and 2 built a unit that answers 1 or 0 and checked it against a truth table. We now turn to a real dataset, recorded a few miles from campus, and a unit whose output estimates the *probability* of a class.

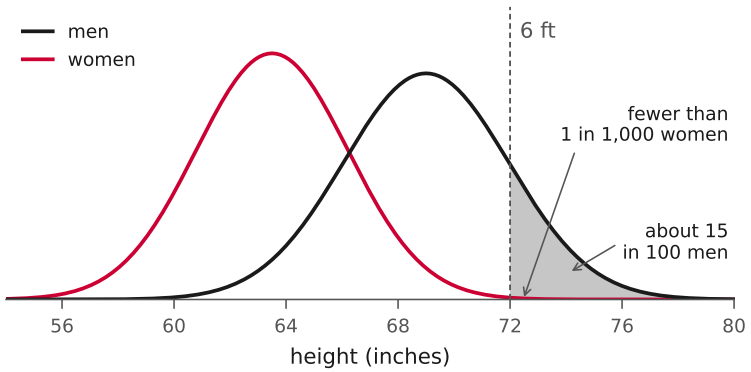
Act 1. Uncertainty in Measurements

1.1 Six feet tall

Two people are each six feet tall. One was born in 1950, the other in 2000. From height alone, how confidently could you predict whether each person is male or female? In class the question was phrased as "XX or XY?" The height survey below records sex, not chromosomes, so its estimates concern the male and female categories in that survey.

Height alone does not determine the survey's sex category. Women and men can both be six feet tall. We can estimate a probability from height data for a specified population; estimates proposed in class ranged from about 50 to 89 percent.

Measurements let us estimate that probability for a particular population.



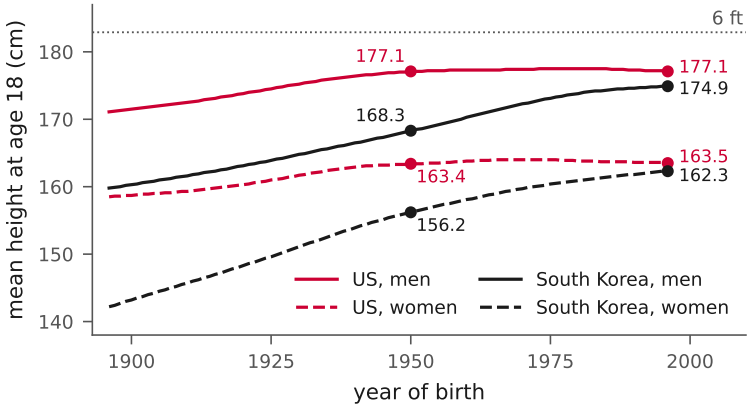
The curves approximate adult height distributions in the United States — women on the left, men on the right — using a national survey that measured heights. In these normal approximations, about **15 in every 100 men** reach six feet, compared with **fewer than 1 in 1,000 women**. These are areas under the curves to the right of 72 inches.

For someone whose height is *near* 72 inches, compare the heights of the two curves there instead. Their ratio is about 75 to 1. Assuming equal numbers of men and women in the population we sample, this gives $75/(75+1) \approx \mathbf{0.99}$ for male. At $5'9''$ the same calculation gives about 0.88. These are estimates from a model; they depend on the population, its proportions, and how well the curves describe it.

The curves: NHANES 2015–2018 (Fryar CD, Carroll MD, Gu Q, Afful J, Ogden CL, *Anthropometric reference data for children and adults: United States, 2015–2018*, Vital Health Stat 3(46), 2021), adults 20 and over; drawn as normal approximations to the survey's percentiles.

1.2 Does the year of birth change it?

Changes in nutrition and health care can affect height distributions across birth cohorts. The size of such changes needs to be checked in the relevant data. The United States data show little change in mean height between these cohorts. Means alone, however, do not tell us whether the probabilities are unchanged.



The plot shows mean height at age 18 by year of birth, over a century. In the United States, men born in 1950 and 1996 both average about 177 cm; the corresponding averages for women are 163.4 and 163.5 cm. The growth happened earlier, then largely levelled off. In South Korea the same plot is still climbing: women born in the 1990s are about 6 cm taller than women born in 1950, and men about 7 cm taller. Populations can move the other way too. A family example came up in class: in the instructor's family in Greece, those who grew up during and just after the war of the 1940s were shorter than their parents, following years of poor nutrition. That is a family recollection, not an estimate for the whole country.

The probability depends on the population's height distributions and class proportions. These can vary with place and time, so data from one population may give a poor estimate for another.

NCD Risk Factor Collaboration, *A century of trends in adult human height*, eLife 5:e13410 (2016) — mean height at age 18 by year of birth, 1896–1996; the last cohort in the data is 1996, which is why the plot and the text say "the 1990s."

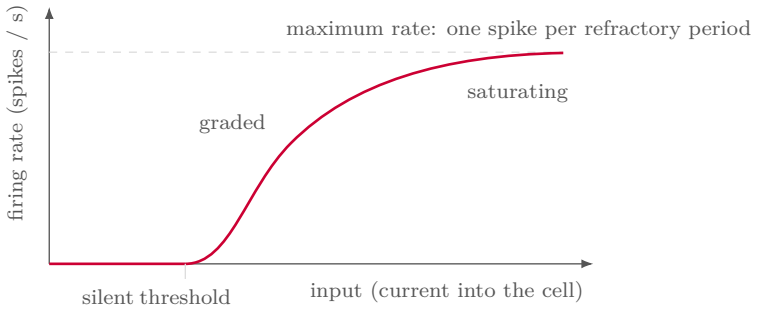
1.3 Uncertainty in a prediction

A gate's truth table specifies one output for every input. Height measurements are continuous, subject to measurement error, and compatible with either survey category. A probability expresses the uncertainty that remains when predicting from height alone.

You make judgments like this all day: you can choose an answer while remaining unsure. Your hesitation on an ambiguous case is a familiar way to think about confidence, without assuming that the brain literally calculates the probability in our example.

1.4 Variation in firing rate

The threshold model in meeting 1 omits variation in a neuron's firing rate. The following schematic shows how that rate can depend on input.



In this schematic, below its threshold the cell is silent. Above it, the cell fires spikes at a *rate* that rises with the input. At high input levels the rate saturates. A cell has only so much energy, and after each spike it needs a recovery interval, the refractory period, before it can fire again. This limits its firing rate; the rates and recovery times differ by cell type. The response drawn here varies continuously within a bounded range. Our artificial unit can also have a graded output, which we will use to represent a class probability. This use of a graded output is an analogy; biological firing rates have a different interpretation.

The firing-rate curve is a schematic; the axes carry no numbers because the rates differ by cell type by an order of magnitude. The standard treatment is P. Dayan and L. F. Abbott, *Theoretical Neuroscience* (MIT Press, 2001), chapter 5.

Act 2. The Newark Weather Dataset

2.1 A day of weather, as numbers

The dataset is the weather at Newark Liberty airport, one row per day from 2006 through 2024, taken from the record NOAA keeps for the station. Four numbers describe a day: the **high** and the **low** temperature, the average **humidity**, and the average sea-level **pressure**. These four features represent each day as a point in four-dimensional space. We can plot selected features, but an ordinary scatter plot cannot show all four dimensions as separate spatial axes.

There are 6,940 rows. Each carries a **label**: 1 if the day falls in the *warm half* of the year, May through October, and 0 if it falls in the *cold half*, November through April. The label is computed from the date, and for that reason the date is not one of the inputs — §4.3 explains how including it would reveal the label. The question the unit is asked: from the four measurements alone, which half of the year is it?

2.2 A notebook, run top to bottom

The meeting's data work lives in a notebook, *The DS Excursion*, which you can run yourself: it is linked from Canvas, and it opens directly in Colab — where you can run it, change it, and keep a copy — from this address:

https://colab.research.google.com/github/ikoutis/ikoutis.github.io/blob/master/course-notes/artificial-neuron-notes/notebooks/the-ds-excursion.ipynb

The notebook loads, checks, and plots the data before applying a classifier.

The data is stored in a public CSV file with one row per day. The following code loads it and displays basic summaries:

```
import pandas as pd

URL = ("https://ikoutis.github.io/course-notes/"
       "artificial-neuron-notes/data/newark-weather.csv")
df = pd.read_csv(URL, parse_dates=["date"])

df.head()
df.shape
df.describe()
```

The library is **pandas**, and `pd` is its customary short name. What `read_csv` returns is a **DataFrame**: a table held in memory, one row per record and one named column per quantity, so that Python sees the spreadsheet the way you do. `head` shows the first rows; `shape` counts rows and columns; `describe` gives summary statistics such as means, spread, and smallest and largest values. The four measurements are the table's **features**: the columns a classifier is given to work with.

Nobody is asked to write these lines from memory. It was said plainly in class that the instructor would need a moment to look them up too, and that an AI assistant can help write this kind of code. What you are asked to know is what each line *does*, and to check that it does what you intended.

2.3 Missing measurements

Check for missing values before analyzing the measurements.

```
df.isna().sum()

FEATURES = ["tmax_f", "tmin_f", "humidity_pct", "pressure_hpa"]
df = df.dropna(subset=FEATURES)
```

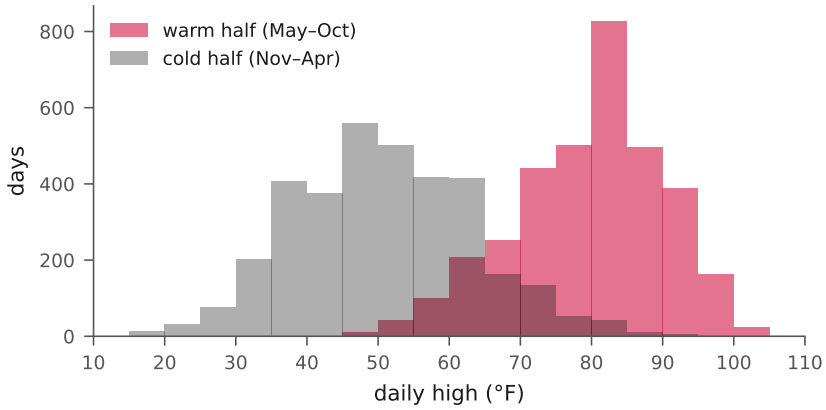
Of the 6,940 days, **64 have no humidity reading and 75 have no pressure reading**. Instruments can fail or stop reporting, leaving gaps in the record. Missing values alone do not tell us which of these things happened. The same problem occurs with factory sensors and many other sources of data.

Three common approaches were discussed in class. You can **drop** the row, leaving out an incomplete record. You can **fill** the hole with the column's average — put in an estimate rather than nothing. Or you can fill it more carefully from *similar* days: the 64 days without a humidity all have a temperature, so take the days with the same temperature and use their average humidity. The notebook drops incomplete rows, leaving **6,865 days — 3,455 in the warm half and 3,410 in the cold half**, almost exactly balanced.

Check for missing values before fitting a model. Dropping rows is simple, but it can bias the remaining sample if the gaps occur more often under particular conditions.

2.4 One measurement, two classes

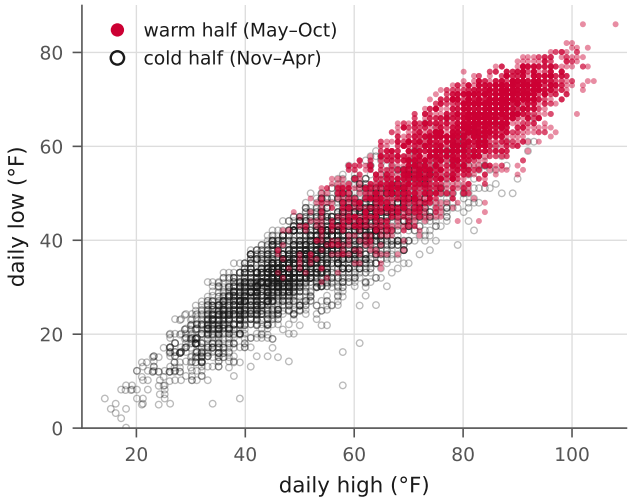
Part of every data-science workflow is looking at the data before asking anything of it, and there is a course devoted to nothing but that (DS 450, data visualization). Here two pictures are enough. The first is a **histogram** of the daily high.



Read it as a count: the horizontal axis is the day's high, in bins of five degrees, and the height of a bar is the number of days whose high fell in that bin — a handful above 100°F, a handful near 20, and the bulk in between, in the bell shape that a great many measured quantities take. The two halves of the year are drawn as two overlaid histograms. Below about 55°F nearly every day is cold-half; above about 75, nearly every day is warm-half; and between 55 and 75 the two shapes overlap — a day with a high of 65 is warm-half about as often as not. Look also at the far right of the gray histogram: cold-half days with a high of 80°F and more. Most are in April, but three are November days in this record. (A prediction was ventured in class that there would be more this November. Treat it as a prediction.)

2.5 Two measurements

The second picture is a **scatter plot**: two measurements at once.



Each day is one point — its high on the horizontal axis, its low on the vertical — filled red for the warm half and hollow for the cold. Set this beside the unit square of meeting 2. That square had two inputs and four corners;

this plane has two inputs and 6,865 points. Every day has a label, just as every corner did. The difference is that the two clouds overlap: the high and low temperatures do not uniquely determine that label.

A histogram uses one axis for a measured variable and the other for counts. A two-dimensional scatter plot uses both axes for measurements; three measurements can be shown in a three-dimensional scatter plot. With more features, selected two- or three-dimensional views can still help, but each leaves out some information.

Example 2.1 Mark the surprising days

Look at the scatter plot and find three days that seem to sit in the wrong cloud. In class two kinds were pointed out on the projected plot. Up in the red cloud, among the hot days, sit a few hollow points: cold-half days whose high and low both belong to summer. Down in the gray cloud sit a few red points: warm-half days on which both the high and the low were winter temperatures. Before deciding that either is an error, write down what you would want to know about it.

Plots can reveal unusual observations or small clusters that summary statistics conceal. Knowledge of the data source helps determine which patterns deserve further investigation.

2.6 The 80° day in November

Here are four surprising days, looked up by date. Measurements in this table are rounded to whole numbers.

date	high	low	humidity	pressure	label
2022-11-07	81°F	56°F	45%	1022 hPa	cold half
2024-11-01	82°F	58°F	47%	1015 hPa	cold half
2024-11-06	83°F	63°F	57%	1016 hPa	cold half
2020-05-09	50°F	34°F	44%	1012 hPa	warm half

The four numbers of a day live on very different scales — temperatures in tens of degrees, humidity a percentage, pressure a number near a thousand — and that is simply what the instruments record. Look instead at the labels. **Every one of them is correct.** The high on November 6, 2024 was 83°F, and it was November; the high on May 9, 2020 was 50°F, with a low of 34, and it was May. These records describe unusual weather; the label does not need correcting just because the day was surprising. The May day fell in the first spring of the pandemic, a period that people in class remembered. The label records which half of the year a day belongs to — not what the weather was like.

November 6, 2024 looks much like a summer day in these measurements. That makes it difficult to classify from weather alone. Similarity is not, by itself, a proof that no rule could classify every recorded day correctly; identical inputs with different labels would establish that.

2.7 Properties of observational data

The examples illustrate three practical issues:

- **The same input can carry either label.** Two six-foot-tall people born in the same year: identical data, and yet they may have different labels. An 83°F high is an ordinary June day, and it happened in

November. A true story from class: the instructor's mother's pet weighs 16 pounds — cat or dog? The room said dog. It is a cat.

- **Measurements have limits.** Instruments have finite precision and can produce errors or missing readings.
- **There are more rows than you can look at.** A gate had four; this table has 6,865; the records of one hospital system for one year run to millions.

These examples motivate an answer that expresses uncertainty. The unit should be able to say which class it favors and how confident it is, even when the measurements do not settle the answer.

Act 3. The Sigmoid Function

3.1 What is a probability?

A probability lies between 0 and 1. We interpret predictions in terms of the frequencies expected across repeated observations from a specified population. If you were handed many people fitting the description "near six feet tall," under the population assumptions of §1.1, the estimate says about 99 percent would be male. A model reporting 0.99 is making that kind of claim; whether its probabilities match observed frequencies must be checked. For the 16-pound pet example, we would need data before attaching a number to "probably a dog."

On the scatter plot the same reading is available directly. Take any one point in the plane and collect all the days that land near it; one estimate of the probability of "warm half" is the *fraction* of those nearby days that are warm-half. For the four corners of a truth table that fraction is always 0 or 1 — each row has exactly one answer. In the weather sample that fraction is often near 0 or 1 far from the band, and somewhere in between inside it. A fraction of 1 in a finite sample does not guarantee that every future day will have that label.

3.2 The score is unbounded

Return to the unit of meeting 2. Its score is

$$s \;=\; \mathbf{w} \cdot \mathbf{x} + b,$$

the weights times the inputs, plus the bias (which is the threshold moved across: $b = -\theta$). The main calculation is still just multiplication and addition. Now look at what the score *ranges* over. The weights can be any numbers and the inputs can be any numbers, so the score runs from $-\infty$ to $+\infty$; on the weather, with the line drawn later in this meeting, it runs from about -100 to about $+75$. But a probability lives in $[0, 1]$. What is needed is a function that takes a score, anywhere on the number line, and returns a probability — and it is clear what the function has to do:

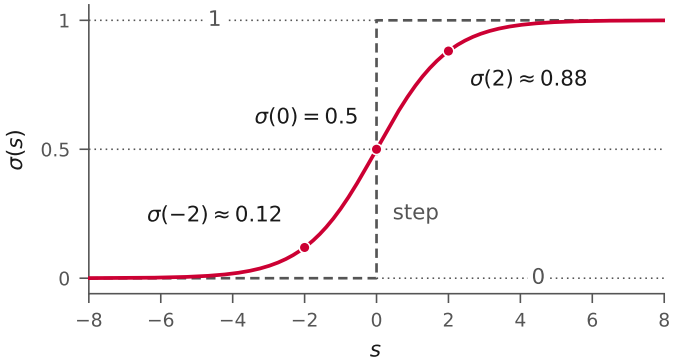
- it should be **increasing**: a low score means a low probability near zero; a high score means a high probability near one;
- at score 0, it should give $\frac{1}{2}$;
- it should be **smooth**, with no jump, and it should **never quite reach** 0 or 1.

3.3 The sigmoid

A function with these properties is the **sigmoid**:

$$\sigma(s) \;=\; \frac{1}{1+e^{-s}}.$$

Take the score, flip its sign, put it in the exponent, and form the ratio. Its graph is the red curve:



Very negative scores give probabilities near 0; very positive ones give probabilities near 1; in between there is a smooth transition that passes through exactly $\frac{1}{2}$ at $s = 0$. The name *sigmoid* means S-shaped. Three useful properties are:

- $\sigma(0) = \frac{1}{2}$;
- $\sigma(-s) \;=\; 1 - \sigma(s)$: the curve is symmetric about the middle point, so a score and its negative give complementary probabilities;
- the limits 0 and 1 are never reached.

The dashed curve is the threshold function from meeting 1. The sigmoid replaces its jump with a smooth transition, producing values that can be interpreted as predicted probabilities. It recalls the graded response of a biological neuron, while serving a different purpose: estimating a class probability.

Example 3.1 Sigma by hand

Compute $\sigma(s)$ for four scores, using $e^2 \approx 7.39$ and $e^{10} \approx 22026$.

For $s = 0$: the exponential of 0 is 1, so $\sigma(0) = 1/(1+1) = 0.5$. This is the value at the decision boundary.

For $s = 2$: a common sign error is to put $e^2 \approx 7.39$ into the denominator as it stands, $1/(1+7.39) \approx 0.12$. This would give a probability below 0.5 despite the positive score. The exponent is $-s$, so the denominator uses $e^{-2} = 1/e^2 \approx 1/7.39 \approx 0.135$. Then

$$\sigma(2) \;\approx\; \frac{1}{1+0.135} \;\approx\; 0.88.$$

Thus the model assigns about 88 percent probability to class 1.

For $s = -2$: no new arithmetic. By the symmetry, $\sigma(-2) = 1 - \sigma(2) \approx 0.12$. The probability assigned to class 0 is about 0.88.

For $s = 10$: $e^{-10} \approx 1/22026 \approx 0.000045$, so $\sigma(10) \approx 0.99995$, which is close to 1 but below it.

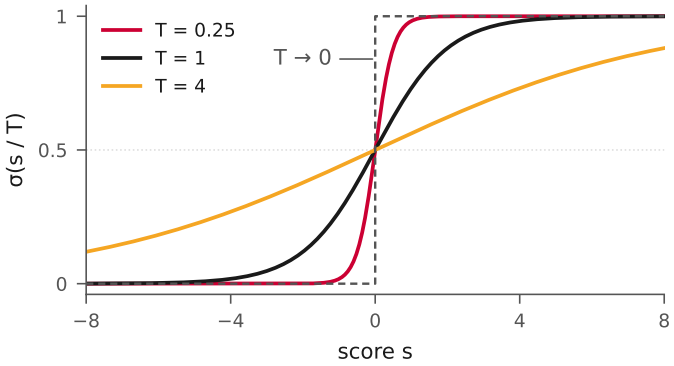
Two questions follow. *For which s is $\sigma(s) \geq \frac{1}{2}$?* For $s \geq 0$ — a student gave the answer at once, and it is the content of §3.5. *Make the sigmoid infinitely steep; which function is left?* Squeeze the transition until it happens all at once, and away from $s = 0$ the limit is the step of meeting 1. At $s = 0$, however, the sigmoid stays at $\frac{1}{2}$; our threshold unit assigns 1 there.

3.4 Temperature

A positive **temperature** T rescales the score before the sigmoid:

$$\sigma\left(\frac{s}{T}\right).$$

Think about what dividing by 10 does: zero stays zero, and every other score moves ten times closer to zero. The transition therefore spreads out — a larger score is needed to produce the same high probability. Divide by $\frac{1}{4}$ instead, which is multiplying by 4, and the transition sharpens.



A small T makes the curve steeper and closer to the step: the unit becomes more confident; away from score zero, the limit is the step. A large T flattens the curve toward $\frac{1}{2}$ everywhere: its outputs move closer to 0.5. A probability of $\frac{1}{2}$ expresses the same uncertainty as a fair coin; computing it does not actually flip a coin. On the weather, where the score is in degrees Fahrenheit, T is a *width in degrees*: how far from the boundary a day has to be before its probability approaches 0 or 1.

The name is **temperature**, and a student recognized it from large language models: the interface to one exposes a parameter called the temperature, and it is this same T, applied to the model's scores for its next token. Low temperature concentrates probability on the most likely options; higher temperature spreads it more evenly and makes sampling more varied — more "poetic," as it was put in class. It does not guarantee accuracy or creativity. The choice of temperature controls how concentrated the output distribution is. The word itself comes from physics, where the same formula appears with an actual temperature in it.

OpenAI, *On the Navier–Stokes Millennium Prize Problem* (September 8, 2026). A September 10 update states that OpenAI's investigation found Buckmaster's Codex prompts could not have influenced its system. That is the company's account.

3.5 The decision boundary

A probability can be converted into a class decision. For example, a 0.5 cutoff assigns "cat" to a prediction of 0.51 for that class. In class it was put in terms of betting: a 51-percent chance of winning gives a positive expected

gain if a win and a loss have equal payoffs and the probability is accurate. It does not guarantee a profit over a finite series. Confidence can be converted into a decision by **thresholding** it at $\frac{1}{2}$ to obtain a binary label.

Thresholding at $\frac{1}{2}$ gives the following equivalence for every $T > 0$:

$$\sigma\left(\frac{\mathbf{w}\cdot\mathbf{x}+b}{T}\right) \geq \tfrac{1}{2} \iff \frac{\mathbf{w}\cdot\mathbf{x}+b}{T} \geq 0 \iff \mathbf{w}\cdot\mathbf{x}+b \geq 0.$$

The first step is the sigmoid's own property — it is at or above $\frac{1}{2}$ exactly when its argument is at or above 0 — and the second is that dividing by a positive number does not change a sign. So thresholding the probability at $\frac{1}{2}$ is *identical* to thresholding the score at 0, whatever T is: the set of inputs the unit calls "more likely warm than not" is precisely the half-plane meeting 2 drew. The predicted probabilities vary with T, while this boundary remains fixed.

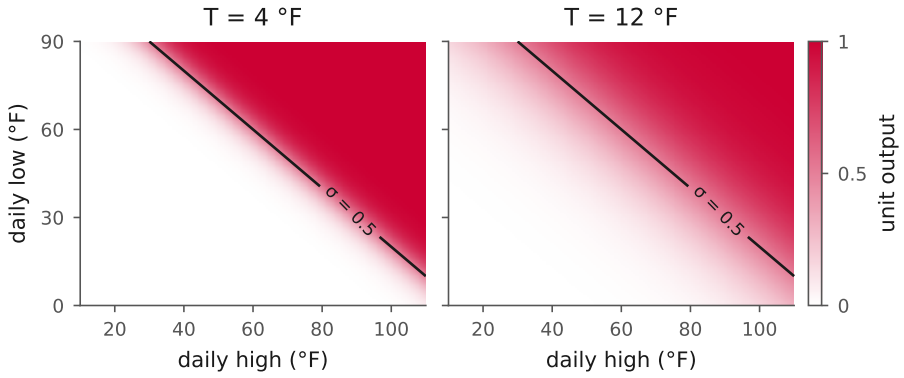
There is a second way to say the same thing. Write P for the unit's output and solve for the score: $P/(1-P)$ is the **odds**, and its logarithm, the **log-odds**, is

$$\log\frac{P}{1-P} = \frac{\mathbf{w}\cdot\mathbf{x}+b}{T},$$

which is linear in $\mathbf{x}$. Underneath the S-shaped curve, the unit is still doing what it did in meeting 2: computing a line.

3.6 Probability contours

To see how the predictions change, color the plane by the unit's probability output instead of its class decision.



A unit that answered with certainty would paint the plane in two flat colors, red on one side of the line and white on the other. This unit paints in *shades*. At the boundary, the predicted probability is exactly 50 percent. Just on the warm side it says 51, 52, 53 percent; move farther away from it and the shade deepens quickly toward one on the warm side and fades toward zero on the cold. The two panels use two temperatures, 4°F and 12°F, and the 0.5 contour is the same line in both. T controls the width of the probability transition around that boundary.

3.7 Comparing the two units

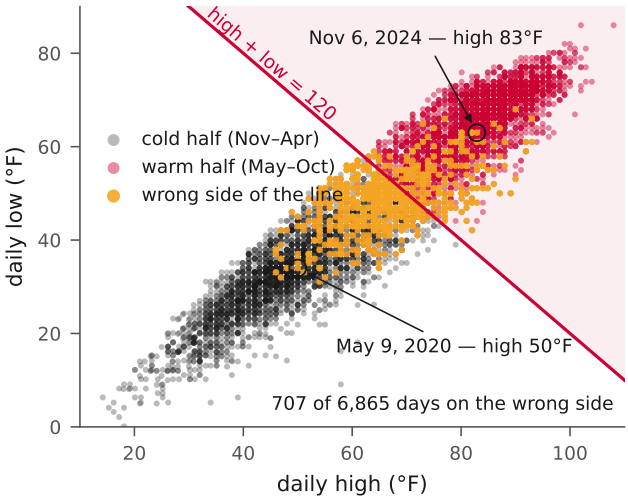
	meeting 1's unit	today's unit
output	$\{0, 1\}$	$(0, 1)$, strictly between
boundary	a line	the same line
settings	$\mathbf{w}$, b	$\mathbf{w}$, b, and one more number, T

Why "strictly between"? A student gave the reason in class: the sigmoid never reaches 0 or 1, whatever the score. In exact arithmetic, finite scores give outputs strictly between 0 and 1. Floating-point arithmetic has finite precision, and a large enough score produces an output that rounds to exactly 1.0 or exactly 0.0. This explains why a notebook can print a probability of 1.0 for something that is not certain.

Act 4. Interpreting Classification Errors

4.1 A line by hand

Back to the data, with the new unit in hand. Draw a line through the band, by eye: where the high and the low add up to 120 — that is, where their average is 60°F — and call the side above it warm.



That line puts **89.7 percent** of the days on the correct side. The other **707 days**, of 6,865, are drawn in gold: they sit on the wrong side of the line. The predicted class is incorrect on these days. The predicted probabilities also indicate how strongly the model favored that class; §4.2 examines one example. A fitted line reaches about **91 percent** on these same days. That is a result, not a proof of the best possible accuracy. We can rule out 100 percent from high and low alone for a different reason: some identical (high, low) pairs have different labels.

4.2 Why not 100 percent

Inside the band the same (high, low) occurs in both halves of the year, and a line gives each point one side. That much is geometry. The overlap reflects the calendar. **April and October have overlapping weather but different labels**: the mean daily high is 63°F in April and 68°F in October, and the label calls one cold and the other warm by the month, not by the weather. Ask which days the line gets wrong and the answer is mostly October, May, and April days — which helps explain the unit's errors. Producing such explanations is a discipline of its own now, explainable AI: looking at a model's answers and saying why it gave them. Overlapping measurements across the two calendar classes explain these errors without requiring measurement faults.

Example 4.1 The probability assigned to an incorrect prediction

Take the day of the table in §2.6, November 6, 2024: a high of 82.9°F and a low of 63.0. Under the line of §4.1 its score is

$$s \;=\; 82.9 + 63.0 - 120 \;=\; 25.9,$$

well above zero. Run it through the sigmoid with a temperature of $T = 10$:

$$\sigma\!\left(\frac{25.9}{10}\right) \;=\; \sigma(2.59) \;\approx\; 0.93.$$

The unit says: *93 percent warm half.* The day is in the cold half. So the unit is wrong — and it gave 7 percent to what actually happened. The threshold unit would return only the class label. A wrong answer with a probability attached also tells us how strongly the unit favored the wrong class. Keep track of the 7 percent: it is the probability this model assigned to the outcome that occurred.

4.3 When the answer was hidden in the data

An often-repeated story about early neural networks was discussed in class. Its original source has not been verified. A network was trained to find tanks in photographs and seemed to do it perfectly — until it turned out that every photo with a tank in it had been taken in one kind of weather, and every photo without one in another. The network had learned to tell sunny days from cloudy days. The model had used a shortcut: weather happened to correlate with the labels in the training photographs.

Documented examples of models using unintended correlations include:

- A network trained to read chest X-rays could tell, from the film alone, *which hospital* it came from — and since hospitals differ in how sick their patients are, the hospital leaked into the diagnosis.
- A classifier recognizing horses in photographs relied on a photographer's *tag* in the corner of many horse images.
- A network rating skin lesions rated them more malignant when purple *surgical markings* were visible in the picture.

These are examples of *shortcut learning*: a model relies on a correlation that may not hold when it is used elsewhere. A related problem is *data leakage*, when the model receives information that reveals the answer or would not be available for the intended task.

The Newark data has an obvious opportunity for leakage. The label is computed from the date. A model allowed to use the date could recover the label from the month without learning anything about weather. That is why the date is excluded from the inputs and used only to look days up. Evaluating this classifier requires checking both its inputs and its prediction accuracy.

The tank story: G. Branwen, *The Neural Net Tank Urban Legend*, https://gwern.net/tank — tellings traced to the early 1960s, no documented failure. The documented cases: Zech et al., *PLOS Medicine* 15(11): e1002683 (2018); Lapuschkin et al., *Unmasking Clever Hans predictors and assessing what machines really learn*, Nature Communications 10:1096 (2019); Winkler et al., *JAMA Dermatology* 155(10) (2019).

4.4 Review

- The hand-drawn line gets about nine days in ten right; a fitted line does a little better. Neither result establishes the best possible model. However, identical (high, low) pairs with different labels make perfect classification impossible from those two features alone.
- The unit outputs a predicted probability and has a linear boundary at a 0.5 cutoff. The parameters in these examples were chosen by hand; learning them requires a search procedure.

Act 5. Practice

5.1 Scores and sigmoid outputs

Matching. Match each score condition to the corresponding sigmoid output.

Score conditions: $s < 0$; $s = 0$; $s > 0$.

Output conditions: $q > 1/2$; $q < 1/2$; $q = 1/2$.

5.2 Assessing probability estimates

Short explanation. A classifier assigns probability 0.9 to class 1, but the observed label is 0. Does this one observation establish that its probability estimates are unreliable? What evidence would help you assess predictions near 0.9?

5.3 Temperature and predicted classes

True or false, with explanation. For fixed scores, increasing the temperature in

$$q = \sigma(s/T), \qquad T > 0,$$

can change the predicted classes when the classification threshold is $1/2$. Also describe how the probabilities change.

5.4 Using the month as a feature

Application. The task is to infer the season label from weather measurements. The labels are defined by calendar month. A student adds the month as an input and obtains almost perfect accuracy. What information is the model using, and why does this result tell us little about its ability to infer the label from weather?